

Invariant points and lines

Basic skills

Q1. $M = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix}$

a) $M \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 12 \end{pmatrix}$

b) $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x+4y \\ 5x-2y \end{pmatrix}$

c) For $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{aligned} 3x+4y &= 1 \\ 5x-2y &= 1 \end{aligned}$

so $\begin{aligned} 10x-4y &= 2 \\ 3x+4y &= 1 \end{aligned} \Rightarrow 13x = 3 \Rightarrow x = \frac{3}{13}$
and $y = \frac{5x-1}{2} = \frac{1}{13}$

so $\frac{1}{13} \begin{pmatrix} 3 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ under M

Q2. $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix} \quad \begin{pmatrix} -3 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3x+2y \\ y \end{pmatrix}$

so $x' = -3x+2y \quad y' = y$

Q3. As $M \underline{0} = \underline{0}$, origin is invariant



Q4. $\begin{pmatrix} -3 & 4 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ so $(1,1)$ is invariant

For all

$$\begin{pmatrix} -3 & 4 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} -3x + 4y &= x \\ 3x - 2y &= y \end{aligned}$$

so $x=y$ so all x s.t. $\begin{pmatrix} x \\ x \end{pmatrix}$
e.g. $\begin{pmatrix} -2 \\ -2 \end{pmatrix}, \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \dots$

Q5. $\begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 3x - y &= x \\ 2x &= y \end{aligned}$

so $y=2x$ is invariant points so
any point $\begin{pmatrix} x \\ 2x \end{pmatrix}$ for x

Yes $y=2x$ is an invariant line

Q6. $\begin{pmatrix} 4 & 2 \\ 6 & 5 \end{pmatrix} \begin{pmatrix} x \\ 2x-3 \end{pmatrix} = \begin{pmatrix} 4x + 4x - 6 \\ 6x + 10x - 15 \end{pmatrix}$
 $= \begin{pmatrix} 8x - 6 \\ 16x - 15 \end{pmatrix}$

now $x' = 8x - 6$ $y' = 16x - 15$

$\Rightarrow y' = 2x' - 3$ so $y = 2x - 3$ is an invariant line

Q7.

A line of invariant points is an invariant line. However an invariant line is not a line of invariant points

Q8. $\begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ mx + c \end{pmatrix} = \begin{pmatrix} 4x - mx - c \\ -2x + mx + c \end{pmatrix}$

so $x' = (4-m)x - c$, $y' = (m-2)x + c$
and if invariant $y' = mx + c$

$$\text{so } (m-2)x + c = m((4-m)x - c) + c$$

$$\Rightarrow (m-2)x + c = (4m-m^2)x + c(1-m)$$

$$\Rightarrow (m^2-3m-2)x + mc = 0$$

As must be true for all x :

$$m^2-3m-2=0 \text{ and } cm=0$$

so $m = \frac{3 \pm \sqrt{9+8}}{2}$ and $c=0$ ↖ As $m \neq 0$

so lines have $m = \frac{3 \pm \sqrt{17}}{2}, c=0$

$$y = \left(\frac{3 \pm \sqrt{17}}{2} \right) x$$

Advanced skills

Q9. $M = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$

a) $\begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 5x + 2y &= x \\ 2x + 5y &= y \end{aligned}$

$\Rightarrow \begin{aligned} 4x + 2y &= 0 \\ 2x + 4y &= 0 \end{aligned} \Rightarrow x=0, y=0$

so only the origin

b) $\begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ mx+c \end{pmatrix} = \begin{pmatrix} 5x + 2mx + 2c \\ 2x + 5mx + 5c \end{pmatrix}$

so $x' = (5+2m)x + 2c, y' = (2+5m)x + 5c$

for invariant line $y' = mx' + c$

$$\text{So } (2+5m)x + 5c = m((5+2m)x + 2c) + c$$

$$\Rightarrow (2+5m)x + 5c = (5m+2m^2)x + (2m+1)c$$

$$\Rightarrow (2m^2-2)x + (2m-4)c$$

As for all x :

$$2m^2-2=0, \quad (2m-4)c=0$$

$$\Rightarrow m = \pm 1, \quad c=0$$

$$\text{So } y = \pm x$$

Q10.

$$M = \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

formula $\rightarrow$

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\text{So } \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} x-y \\ x+y \end{pmatrix}$$

For Point need $\frac{\sqrt{2}}{2} (x-y) = x, \frac{\sqrt{2}}{2} (x+y) = y$

$$\Rightarrow \left(\frac{\sqrt{2}}{2}-1\right)x - \frac{\sqrt{2}}{2}y = 0, \quad \frac{\sqrt{2}}{2}x + \left(\frac{\sqrt{2}}{2}-1\right)y = 0$$

$$\Rightarrow y = \left(1 - \frac{2}{\sqrt{2}}\right)x, \quad y = \frac{\frac{\sqrt{2}}{2}x}{1 - \frac{\sqrt{2}}{2}}$$

← rationalise

$$y = \frac{\sqrt{2}x}{2 - \sqrt{2}} \times \frac{2 + \sqrt{2}}{2 + \sqrt{2}} = \frac{(2\sqrt{2} + 2)x}{2} = (\sqrt{2} + 1)x$$

and as $1 - \frac{2}{\sqrt{2}} \neq \sqrt{2} + 1$ no invariant points

For lines let $y = mx + c$ then

$$x' = \frac{\sqrt{2}}{2}(x - y) = \frac{\sqrt{2}}{2}(x - (mx + c))$$

$$y' = \frac{\sqrt{2}}{2}(x + y) = \frac{\sqrt{2}}{2}(x + mx + c)$$

with $y' = mx' + c$

$$\Rightarrow \frac{\sqrt{2}}{2}((m+1)x + c) = \frac{\sqrt{2}}{2}((m - m^2)x + cm) + c$$

$$\Rightarrow (m+1)x + c = (m - m^2)x + cm + \frac{2}{\sqrt{2}}c$$

$$\Rightarrow (m^2 + 1)x + \left(1 - \frac{2}{\sqrt{2}} - m\right)c = 0$$

as $m^2 + 1 \neq 0$ no invariant lines.

Q11. $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ mx+e \end{pmatrix}$

So $x' = ax + bmx + be$

$y' = cx + dmx + de$

For $y' = mx' + e$

$\Rightarrow (c + dm)x + de = (am + bm^2)x + bem + e$

$\Rightarrow (bm^2 + (a - d)m - c)x + (bm + 1 - d)e = 0$

so

a) For no lines, no real solution to

$$bm^2 + (a - d)m - c = 0$$

$$\Rightarrow (a - d)^2 + 4bc < 0$$

b) For infinitely many must be true
for all m that $bm^2 + (a - d)m - c = 0$

so $b = 0, a = d, c = 0$ e.g. $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$